INCH - POUND

MIL-A-47240B(MI)
27 July 1990
SUPERSEDING
MIL-A-47240A (MI)
28 November 1979

MILITARY SPECIFICATION

AMMONIUM NITRATE (UNCOATED)

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of uncoated ammonium nitrate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 128 - Analysis of Lubricating Grease, Standard Methods of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Chemical analysis. The chemical analysis of the material shall be as specified in table I.

TABLE I. Chemical analysis.

Composition	Minimum	Maximum
Ammonium nitrate (NH_4NO_3), percent	99.0	-
Moisture, percent	-	0.50
Ash, percent	-	0.10
Ether-soluble material, percent	-	0.10
Water insolubles, percent	-	0.10
Soluble chlorides, as NH_4C_1, percent	-	0.03
Acidity, as free HNO_3, percent	-	0.02
Acidity, pH	5	7

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3.3 Safety and health. Ammonium nitrate is an oxidizing material. When contaminated with organic materials, it is sensitive and may explode when subjected to friction or fire. The following conditions shall apply:

The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.4 Workmanship. The ammonium nitrate shall be uniform in quality, free from foreign materials, and shall be manufactured under conditions and procedures standard in the industry (see 4.5.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.3.1 Lot formation. A lot shall consist of materials produced by one manufacturer with no change in formulation or process. If manufacture is by batch process, each batch shall constitute a lot.

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4.4 First article inspection. When specified in the contract or order (see 6.2), a first article sample shall be subjected to first article inspection. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table II.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
First article	3.1	4.4
Ammonium nitrate	3.2	4.6.1
Moisture	3.2	4.6.2
Ash	3.2	4.6.3
Ether-soluble material	3.2	4.6.4
Water solubles	3.2	4.6.5
Soluble chlorides, as NH_4Cl	3.2	4.6.6
Acidity, as free HNO_3	3.2	4.6.7
Acidity, pH	3.2	4.6.8
Workmanship	3.4	4.5.2

4.5.1 Sampling procedures. Sampling for inspection and testing shall be as specified herein. The number of containers to be selected at random, for sampling shall be equal to the square root of the total number of containers in the lot. If the number obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot, in which case each container shall be sampled). The samples shall be classified as a primary sample and a composite sample as specified herein.

4.5.1.1 Primary sample. From each selected container, a sample shall be taken from three or more places throughout the container. The total weight of the samples taken from each container shall be greater than 50 grams (g). Each sample thus taken shall be mixed thoroughly, placed in a clean dry container, and labeled to identify the material name, original container designation, contract number, and lot number.

4.5.1.2 Composite sample. Each primary sample shall be subdivided at random to prepare a composite sample (shall be not greater than 500 g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample thoroughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designation, contract number and lot number. All specified chemical and physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall result in rejection of the lot represented.

4.5.2 Examination. The ammonium nitrate shall be visually inspected to determine compliance with 3.4.

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4.6.1 Ammonium nitrate (NH_4NO_3). Add 25 milliliters (ml) of approximately 40 percent formaldehyde solution and a few drops of phenolphthalein indicator solution to 100 ml of freshly boiled distilled water in a 125-ml Erlenmeyer flask or equivalent. Neutralize the solution with 0.15N sodium hydroxide (NaOH), add a weighed portion of approximately one gram of sample (weighed to the nearest 0.01g), heat mixture to 60°Celsius (C) (140°Fahrenheit (F); then cool to room temperature. Titrate the solution with 0.5N NaOH to an end point which persists at least 30 seconds. Value obtained for percent of ammonium nitrate shall be not less than specified in 3.2.

$$\text{Percent ammonium nitrate equals } \frac{8.005\,AB - 1.49C}{W}$$

Where: A equals volume of sodium hydroxide used, ml.
B equals normality (N) of sodium hydroxide solution.
C equals ammonium chloride in sample (4.6.6), percent.
W equals weight of sample taken, g.

4.6.2 Moisture determination.

4.6.2.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be Aquameter, Model KF-2 or KF-3, Beckman Instruments, Inc., Fullerton, CA, or approved equivalent. The Aquameter shall be prepared for operation as described in the technical manual furnished by the manufacturer (Beckman Instruments, Inc.). Use of an alternate item of equipment approved by the procuring activity will necessitate use of the specific technical manual prepared by the manufacturer.

4.6.2.2 Reagents.

a. Karl Fischer reagent. Karl Fischer reagent must have a strength such that each milliliter of Karl Fischer reagent corresponds to 0.0014-0.0023 g of water. Dilute 750 ml of commercially available established Karl Fischer reagent (with water equivalent of 0.005-0.007 g/ml) to 2000 ml with absolute methanol (not greater than 0.1 percent water). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

1. Use sodium tartrate dihydrate ($NO_2C_4H_4O_6.2H_2O$) as a primary standard (with a water content of 15.66 percent) for standardizing Karl Fisher reagent. If the water content value is in question, it may be determined by heating some of the salt at 150°C (302°F) for 3 hours. Should the value (as determined) differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; instead of the 15.66 in the formula below, the factor should be 10P, where P is percentage moisture (as determined). Rapidly transfer 0.090-0.110 g weighed to the nearest 0.0001 g) of reagent-grade $Na_2C_4H_4O_6.2H_2O$ to the titration vessel.

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2. Titrate to an end point in the same manner as with the sample (see 4.6.2.3).

3. Repeat the standardization procedure until three successive results agree within five parts per thousand.

4. If the indicated water equivalent (A) of the Karl Fischer reagent is less than 0.0014 g of water per ml Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used. In this case, distill the methanol from metallic calcium or calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, may also reduce the water content of the methanol sufficiently.

Water equivalent (A) equals $\frac{0.1566}{V}$ W

Where: A equals water equivalent of the Karl Fischer reagent, g/ml.
W equals weight of $Na_2C_4H_4O_6 \cdot 2H_2O$ taken, g.
V equals volume of Karl Fischer reagent used, ml.

b. Water-in-methanol solution. The water-in-methanol solution should contain 0.0015-0.0020 g of water per ml of solution. A good grade of commercial absolute methanol contains approximately 0.0010 g of water per ml of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-in-methanol solution to produce a change of 0.0010 g per ml. Determine the relative strength of the water-in-methanol solution in terms of Karl Fischer reagent as follows:

Put about 50 ml of the anhydrous methanol used in 4.6.2.2a into the titration breaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.6.2.2a), then back-titrate with water-in-methanol solution (4.6.2.2b). Then run in an additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back-titrate with water-in-methanol solution (read to the nearest 0.01 ml). Repeat the addition and back-titrating steps twice more to provide triplicate determinations of the equivalency ratio. Calculate the ratio (B) of the Karl Fischer reagent to that of the water-in-methanol solution. The range of the ratios calculated from the three titrations shall be not greater than 0.04. If the range is greater than 0.04, continue making titrations until three ratios are obtained whose range is not greater than 0.04. Then determine the average ratio from all the ratios which have been obtained.

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4.6.2.3 Procedure. Determine the moisture content of the sample to determine compliance with 3.2 by the Karl Fischer method using a direct-titration technique. Introduce the titration beaker, through the opening in the diaphragm, approximately 10 g of sample weighed to the nearest 0.001 g. Close the opening, start the stirrer and press the titrate button to titrate with Karl Fischer reagent (4.6.2.2a). When the indicator light glows, rad the Karl Fischer buret to the nearest 0.01 ml. Where necessary, water methanol solution (4.6.2.2b) shall be used to back titrate.

$$\text{Procedure moisture equals } \frac{100A\,(V_{KF} - BV_{WM})}{W}$$

Where: A equals weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml.

V equals volume of Karl Fischer reagent titrant KF used, ml.
V equals volume of water-in-methanol solution WM titrant used, ml.
B equals ratio of Karl Fischer reagent to that of water-in-methanol solution, ml/ml.
W equals weight of sample taken, g.

4.6.3 Ash. Determine the amount of ash using the procedure given in ASTM D 128 to determine compliance with 3.2.

4.6.4 Ether-soluble material. Ether-soluble material shall be tested to determine compliance with 3.2. Dry a clean, 150-ml beaker in an oven at 100 ± 5°C (212 ± 9°F) for approximately one hour, cool in a desiccator, and weigh to the nearest 0.001 g. Weight to the nearest 0.001 g approximately 25 g of the specimen and transfer to the thimble of a Soxhlet extractor containing anhydrous ether. Extract the specimen for approximately two hours. Save the extracted specimen for the water-insoluble determination (4.6.5). Evaporate the ether extract in the extraction flask to a volume of approximately 50 ml on a steam bath and transfer to the weighed beaker. Rinse out the extraction flask with several 5-ml portions of ether and add them to the main extract in the beaker. Cover the beaker with a ribbed watch glass and evaporate on a steam batch. After the solvent is removed, heat the beaker and residue in an oven at 100°C (212°F) for one hour, cool in a desiccator, and weigh to the nearest 0.001 g.

$$\text{Percent ether-soluble material equals } \frac{100A}{W}$$

Where: A equals weight of residue from sample, g.
W equals weight of sample taken, g.

4.6.5 Water insolubles. Water insolubles shall be tested to determine compliance with 3.2. Using the ether-extracted specimen from 4.6.4, dissolve it in approximately 100 ml of distilled water. Weigh a dry, sintered-glass funnel to the nearest 0.001 g, then filter the mixture carefully through the funnel in order that all undissolved residue may be collected. Dry the funnel and residue for one hour in an oven at 105 ± 5°C (221 ± 9°F), allow to cool to room temperature in a desiccator, then reweigh to the nearest 0.001 g.

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$$\text{Percent water insolubles equals} \quad \frac{(A - B)\ 100}{W}$$

Where: A equals weight of funnel plus residue, g.
B equals weight of funnel empty, g.
W equals weight of sample taken in 4.6.4, g.

4.6.6 Soluble chlorides, as NH_4Cl. The soluble chlorides, as NH_4Cl shall be tested as specified herein to determine compliance with 3.2. Place approximately 50 ml of chloride-free distilled water in a porcelain evaporating dish. Add 5 g of sample (weighed to the nearest 0.0001 g). Acidify the solution with approximately 2 ml of 3N nitric acid. Titrate the solution with contents greater than 0.1N silver nitrate. Add a few drops of ferric alum indicator solution and, while swirling, titrate the excess silver nitrate with 0.1N ammonium thiocyanate. The end point is taken as the first permanent pink or red color observed.

$$\text{Percent ammonium chloride equals} \quad \frac{(AB - CD)\ 5.35}{W}$$

Where: A equals volume of silver nitrate used, ml.
B equals normality of silver nitrate solution.
C equals volume of ammonium thiocyanate solution used, ml.
D equals normality of ammonium thiocyanate solution
W equals weight of sample taken, g.

4.6.7 Acidity, as free HNO_3. Dissolve approximately 100 g of sample (weighed to the nearest 0.01 g) in 400 ml of distilled water. Filter, add methyl red indicator, and titrate with 0.1N sodium hydroxide (NaOH). Calculate the acidity as follows:

$$\text{Acidity, as percent nitric acid equals} \quad \frac{6.3\ N\ V}{W}$$

Where: N equals normality of NaOH solution.
V equals volume of NaOH solution used, ml.
W equals weight of sample taken, g.

4.6.8 Acidity, pH. The acidity, pH, specified in 3.2 shall be determined in accordance with 4.6.8.2.

4.6.8.1 Apparatus. Electrometric pH apparatus, or equal.

4.6.8.2 Procedure. Dissolve approximately one g of sample (weighed to the nearest 0.01 g) in 100 ml of freshly boiled distilled water contained in a 200-ml beaker. Determine the pH of the solution electrometrically using a pH meter.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packing and marking shall be in accordance with the requirements of MIL-STD-1190.

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5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ammonium nitrate covered by this specification is intended for use as an ingredient in ammonium nitrate-base solid propellants.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. When first article is required (see 3.1 and 4.4).

d. First article sample size (see 4.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Wherever inch/pound dimensions are used in this document, metric equivalents in accordance with FED-STD-376 are acceptable.

6.5 Subject term (keyword) listing.

Chemical element
Element, solid propellants
Ingredient, rocket fuel

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-47240B (MI)	2. DOCUMENT DATE (YYMMDD) 27 July 1990

3. DOCUMENT TITLE
AMMONIUM NITRATE (UNCOATED)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDER US ARMY MISSILE COMMAND	(205) 876-6980	746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST Redstone Arsenal, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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